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#Om Bhujbal 22CV007
# Program to calculate the length of transition curve

V = int(input("Enter the value of design speed (kmph): "))
R = int(input("Enter the value of Radius of curvature (m): "))
N = int(input("Enter the value of slope (e.g., 150 for 1 in 150): "))
W = float(input("Enter the width of road including extra widening (m): "))
emax = float(input("Enter the max allowable superelevation (plain terrain): "))

# Calculate actual super elevation
ecal = (V * V) / (225 * R)
print("The value of Super elevation:", ecal)

# Use the minimum of calculated or allowable
if ecal < emax:
    e = ecal
else:
    e = emax

# Length of transition curve
Ls = (e * N * W) / 2
print("The length of transition curve is:", Ls, "m")

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Enter the value of design speed (kmph): 65
Enter the value of Radius of curvature (m): 220
Enter the value of slope (e.g., 150 for 1 in 150): 150
Enter the width of road including extra widening (m): 7.5
Enter the max allowable superelevation (plain terrain): 0.07
The value of Super elevation: 0.08535353535353535
The length of transition curve is: 39.37500000000001 m

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#Om Bhujbal 22CV007
# Program to calculate pavement thickness from EWL and Traffic Index

import numpy as np

R = int(input("Enter Constant R: "))
C = int(input("Enter Constant C: "))

A = int(input("Enter number of EWL Constants: "))
B = int(input("Enter number of AADT values: "))

EWL_Constant = []
AADT = []

for i in range(A):
    val = float(input(f"Enter EWL Constant {i+1}: "))
    EWL_Constant.append(val)

for j in range(B):
    val = float(input(f"Enter AADT {j+1}: "))
    AADT.append(val)

product = np.dot(EWL_Constant, AADT)
print("Total EWL:", product)

EWL_10_years = product * 1.6

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print("EWL after 10 years:", EWL_10_years)

TI = 1.35 * ((EWL_10_years + (product / 2)) ** 0.11)
print("Traffic Index:", TI)

# Pavement Thickness formula
Thickness = 0.166 * TI * ((90 - R) / (C ** 0.2))
print("Pavement Thickness:", Thickness, "cm")

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Enter Constant R: 48
Enter Constant C: 16
Enter number of EWL Constants: 4
Enter number of AADT values: 4
Enter EWL Constant 1: 330
Enter EWL Constant 2: 1070
Enter EWL Constant 3: 2460
Enter EWL Constant 4: 4620
Enter AADT 1: 3750
Enter AADT 2: 470
Enter AADT 3: 320
Enter AADT 4: 120
Total EWL: 3082000.0
EWL after 10 years: 4931200.0
Traffic Index: 7.577910657490486
Pavement Thickness: 30.344701003916345 cm

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#OM Bhujbal 22CV007
# Program to calculate thickness of pavement layers using CBR values

import math

P = float(input("Enter wheel load (kg): "))
p = float(input("Enter tyre pressure (kg/cm^2): "))
M = int(input("Enter total number of pavement layers: "))

pi = 3.14159

for i in range(M):
    CBR_value = float(input(f"Enter CBR (%) for layer {i+1}: "))
    T = math.sqrt((1.75 * P) / CBR_value - (P / (p * pi)))
    print(f"Thickness above this layer: {T:.2f} cm")

print("Note: Given bitumen layer thickness is 4 cm")

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Enter wheel load (kg): 4085
Enter tyre pressure (kg/cm^2): 7
Enter total number of pavement layers: 3
Enter CBR (%) for layer 1: 4.38
Thickness above this layer: 38.03 cm
Enter CBR (%) for layer 2: 6
Thickness above this layer: 31.71 cm
Enter CBR (%) for layer 3: 12
Thickness above this layer: 20.25 cm
Note: Given bitumen layer thickness is 4 cm

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